

SECTION A

Multiple Choice Question

Attempt all questions

(1 × 30 = 30)

1. What is the primary function of the Base Station Controller (BSC) in GSM architecture?
 - a. Voice encryption
 - b. Handoff management between base stations**
 - c. Internet connectivity
 - d. Subscriber billing
2. What does the term "cell splitting" refer to in cellular networks?
 - a. Combining multiple cells into one
 - b. Dividing a cell into smaller cells to increase capacity**
 - c. Changing frequency bands
 - d. Reducing transmission power
3. What is the primary purpose of the D channel in ISDN?
 - a. Voice transmission
 - b. Signaling and control**
 - c. Video streaming
 - d. Internet browsing
4. Which frequency band is commonly used by Wi-Fi networks?
 - a. 900 MHz
 - b. 2.4 GHz
 - c. 5 GHz
 - d. Both B and C**
5. Which component of GSM architecture temporarily stores subscriber information when they roam into a new area?
 - a. HLR
 - b. VLR**
 - c. MSC
 - d. BTS
6. Which duplexing technique, defined by the use of paired channels (forward/reverse) with a fixed frequency gap, is exclusively employed in analog mobile radio systems like AMPS?
 - a. Time Division Duplexing (TDD)
 - b. Half-Duplexing
 - c. Simplex
 - d. Frequency Division Duplexing (FDD)**

7. In a cellular telephone system, which core component acts as the central controller, responsible for managing calls, coordinating handoffs, and interfacing with the Public Switched Telephone Network (PSTN)?
 - a. Mobile Station (MS)
 - b. Base Station (BS)
 - c. Mobile Telephone Switching Office (MTSO)**
 - d. Home Location Register (HLR)
8. Between a pager, a cellular phone, and a cordless phone, which device is expected to have the longest battery life between charges?
 - a. Cellular phone
 - b. Cordless phone
 - c. Pager**
 - d. All of the above
9. If a cellular operator is allocated a total spectrum of 50 MHz and each voice channel occupies 200 kHz, how many simultaneous voice channels can the operator support?
 - a. 125 channels
 - b. 250 channels**
 - c. 500 channels
 - d. 1000 channels
10. Which modulation technique is commonly used in 4G LTE networks?
 - a. AM
 - b. FM
 - c. QPSK
 - d. OFDM**
11. What was the most significant technological shift from 1G to 2G mobile communication systems?
 - a. From frequency hopping to spread spectrum.
 - b. From circuit switching to packet switching.
 - c. From analog modulation to digital modulation.**
 - d. From regional standards to a single global standard.
12. Which 2G standard was primarily adopted across Europe and gained widespread global dominance?
 - a. IS-95 (cdmaOne)
 - b. IS-136 (D-AMPS)
 - c. PDC (Personal Digital Cellular)
 - d. GSM (Global System for Mobile Communications)**

13. The primary limitation of 2G mobile networks that prompted the development of 2.5G enhancements was:
- a. Lack of international roaming capabilities.
 - b. Poor voice call quality and frequent disconnections.
 - c. Low data rates and limited multimedia support.**
 - d. High power consumption for mobile devices.
14. Which 2.5G technology introduced packet-switched data capabilities into GSM networks, enabling services like email and web browsing?
- a. GPRS (General Packet Radio Service)**
 - b. HSCSD (High Speed Circuit Switched Data)
 - c. EDGE (Enhanced Data rates for GSM Evolution)
 - d. IS-95B (cdmaOne + Packet Data)
15. What unique duplexing approach distinguishes China's TD-SCDMA 3G standard from UMTS (W-CDMA) and CDMA2000?
- a. It exclusively uses Frequency Division Duplexing (FDD).
 - b. It employs half-duplex communication for all services.
 - c. It uses Time Division Duplex (TDD) for flexible uplink/downlink switching.**
 - d. It relies solely on simplex communication.
16. In the context of wireless networks, what is the primary characteristic that defines a Personal Area Network (PAN)?
- a. Its short-range communication capability, typically within about 10 metres, for connecting personal devices.**
 - b. Its ability to provide wide-area cellular coverage.
 - c. Its use of Wi-Fi (IEEE 802.11) as the main technology.
 - d. Its focus on fixed wireless access for last-mile connectivity.
17. A GSM system is allocated a total bandwidth of 25 MHz. Given that each GSM carrier occupies 200 kHz and supports 8 time slots (users) per carrier, how many simultaneous users can the system support in total?
- a. 125 users
 - b. 500 users
 - c. 1000 users**
 - d. 2000 users
18. Which of the following was a key goal for the development of Third Generation (3G) wireless networks, as standardized by IMT-2000?
- a. To reduce mobile phone battery consumption compared to 2G.
 - b. To provide multi-megabit internet access, video calling, and 'always-on' access.**
 - c. To enable only voice calls with improved quality over 2G.

- d. To transition all existing analog systems to digital without new infrastructure.
- 19. Which type of wireless system is best suited for providing high-speed broadband data, video, and voice services over short distances (1-5 km) in urban clusters, requiring Line-of-Sight (LOS) communication?
 - a. Wireless Local Area Network (WLAN)
 - b. Cellular Telephone System
 - c. Wireless Local Loop (WLL)
 - d. **Local Multipoint Distribution Service (LMDS)**
- 20. What was the fundamental change in data handling architecture that distinguished 2.5G GPRS/EDGE from traditional 2G GSM systems?
 - a. **The introduction of a packet-switched core network for data traffic.**
 - b. A shift from TDMA to CDMA multiple access.
 - c. Exclusive use of Frequency Division Duplexing (FDD).
 - d. The removal of the Home Location Register (HLR).
- 21. What is the fundamental goal of a cellular system regarding user capacity and spectral efficiency?
 - a. To reduce the number of users while maintaining spectral efficiency.
 - b. To increase spectral efficiency while decreasing user capacity.
 - c. To decrease both user capacity and spectral efficiency for simplicity.
 - d. **To increase user capacity and spectral efficiency.**
- 22. Which entity is responsible for managing call setup, routing, handoffs, and interfacing with the Public Switched Telephone Network (PSTN) in a cellular system?
 - a. Base Station (BS)
 - b. Mobile Station (MS)
 - c. Home Location Register (HLR)
 - d. **Mobile Telephone Switching Office (MTSO)**
- 23. What is the primary purpose of frequency reuse in cellular systems?
 - a. To prevent any signal interference between cells.
 - b. To allow different cells to use entirely separate frequency bands.
 - c. **To allow the same radio channels to be used in non-adjacent cells to improve capacity.**
 - d. To enable mobile devices to switch between analog and digital signals.
- 24. What is the primary disadvantage of FDMA compared to TDMA?
 - a. **Lower spectral efficiency**
 - b. Higher interference
 - c. Complex synchronization

- d. Limited coverage
25. What is the primary purpose of the Guard Band in wireless communication?
- a. Increase data rate
 - b. Prevent interference between adjacent channels**
 - c. Reduce power consumption
 - d. Enhance signal strength
26. Which wireless technology is used for high-speed internet in rural areas?
- a. Wi-Fi
 - b. WiMAX**
 - c. Bluetooth
 - d. Zigbee
27. What was a significant limitation of 1G Analog Cellular Systems (like AMPS) that led to their obsolescence?
- a. They supported high-speed data and multimedia services.
 - b. They offered strong digital encryption for secure communication.
 - c. They utilized advanced CDMA technology for spectral efficiency.
 - d. They had no data support and were susceptible to eavesdropping.**
28. What is the primary benefit of MIMO in wireless communication?
- a. Increased data rates and reliability**
 - b. Lower power consumption
 - c. Reduced interference
 - d. Simpler hardware
29. Which performance criterion measures the probability that a call is blocked due to a lack of available resources during peak hours?
- a. Quality of Service (QoS)
 - b. Grade of Service (GoS)**
 - c. Call Dropping Probability
 - d. Signal-to-Interference Ratio (SIR)
30. Which of the following is a key feature of DECT?
- a. Long-range communication
 - b. Secure voice encryption**
 - c. High data rates
 - d. Cellular roaming

SECTION B

Short Answer Question

Attempt any five (5) questions

(6 × 5 = 30)

1. Discuss the key technological advancements that significantly contributed to the exponential growth of the mobile radio industry. Explain how each advancement enabled the deployment of more complex, efficient, and widespread wireless communication systems **[Chapter 6]**
2. Compare and contrast Frequency Division Duplexing (FDD) and Time Division Duplexing (TDD). Provide an example of a wireless communication system that primarily utilizes each technique and explain the fundamental difference in how they achieve full-duplex communication. **[Chapter 7]**
3. Describe the role of the Mobile Switching Center (MSC) in GSM architecture. **[Chapter 1]**
4. Calculate the traffic intensity (in Erlangs) for a system with 900 calls per hour, each lasting an average of 3 minutes. **[Chapter 6]**
5. Explain the near-far problem in CDMA and how power control mitigates it. **[Chapter 5]**
6. What are the benefits of using microcells in urban wireless networks? **[Chapter 3]**
7. Define Grade of Service (GoS) and explain its importance in cellular network design. **[Chapter 3]**
8. “SIR of 10 dB means the desired signal is 10 times stronger than the interfering signal (on a power basis)”. Explain this statement in details. **[Chapter 4]**

SECTION C

Long Answer Question

Attempt any two (2) questions

(20 × 2 = 40)

1.

- a. A cellular system has a total bandwidth of 30 MHz with 200 kHz channels. If the reuse factor (N) is 4, calculate the number of channels per cell. (10 Marks)
- b. Discuss the key functions of the Home Location Register (HLR) and Visitor Location Register (VLR) in GSM networks. (10 Marks) **[Chapter 4]**

2.

- a. Explain the function of Paging Systems, describing the types of messages they send, their typical coverage areas, and why they need to provide low data rates to achieve better coverage. (5 marks)
- b. Describe Cordless Telephone Systems, including their connection to the Public Switched Telephone Network (PSTN), typical range limitations, and how second-generation systems offered improvements. (5 marks)
- c. Compare and contrast Wireless Local Loop (WLL) and Local Multipoint Distribution Service (LMDS). Your comparison should include their typical frequency bands, primary use cases, coverage ranges, and requirements for Line-of-Sight (LOS). (10 marks) **[Chapter 1 and 2]**

3.

- a. Compare FDMA, TDMA, and CDMA in terms of their suitability for modern wireless networks. Highlight their challenges in ensuring Quality of Service (QoS). (12 Marks)
- b. A GSM system uses a 20 MHz bandwidth with 200 kHz carrier spacing. If each carrier supports 8 time slots, calculate the total number of simultaneous users. (8 Marks) **[Chapter 2, 3 and 5]**